precedent to compare new AI systems against. However, user testing is increasingly necessary for safety evaluations, given the growing concern about potential ethical and social risks from human interaction with generative AI systems – such as overtrust and overreliance; anthropomorphism risks and potential emotional harm; and well-evidenced problems of disparate functionality and externalities between different user groups. AI system and product developing organisations often host user testing expertise and infrastructure. These groups could expand their remit to perform human–AI interaction evaluations to help assess potential risks of harm. As the limitations of capability testing for risk evaluation are becoming clear, more evaluations at this layer may be conducted in the future.

Passive monitoring of human use

The passive monitoring of people's activity on the platform where the AI system is deployed can indicate ethical or social harm. Passive monitoring can be done during pilot releases or after wide-scale deployment. Collected data may reveal effects during an interaction, as well as effects that last beyond the human–AI interaction. For example, prior research on human–AI teaming used a platform where humans could interact with an AI teacher to improve the skill of playing Go. This work identified effects of the human–AI interaction on human skill and playing style that showed after the human had interacted with an AI system (Choi et al., 2022; Shin et al., 2021). Passive monitoring also includes mapping out risk over the course of a "user journey", such as the risk of being exposed to misinformation when seeking information on a search engine (Roy et al., 2023).

For such studies, quasi-experimental data may permit inferences on causal relationships without conducting active interventions. Such work can leverage expert annotation on how human experts interact with AI systems in particular domains or tasks. There are also beginning to be automated tools for evaluating mechanisms in human–AI collaboration, such as an automated approach to detecting patterns of turn-taking in human–AI collaborative writing (Zeng et al., 2023). Passive monitoring can be combined with active interventions for more robust results. For example, one study triangulated reviews from an app store with a follow-up survey to assess problems in user–AI interaction (Eiband et al., 2019). Passive monitoring also lends itself to longitudinal studies, such as assessing transcripts and interaction data to observe human emotional attachment to AI companions and, more broadly, human relationship-building with generative AI systems (Pentina et al., 2023; Xie and Pentina, 2022).

The main limitation of passive monitoring is that it rarely allows the isolation of causal effects, as it makes no direct experimental intervention. To evaluate potential harmful impacts, further experimental testing under controlled conditions may be necessary.

A.2.3. Systemic impact layer

Staged release and pilot studies

Staged release processes or pilot studies can give insight into potential systemic impacts. An AI system can be deployed in a controlled setting. For example, a business, hospital, or public institution may adopt an AI system as a pilot test, with close monitoring to assess impacts such as on the provision of care (Elish and Watkins, 2020). Such monitoring may use ethnographic methods that entail witnessing how a group of people use an AI system in a real-world setting alongside qualitative methods such as interviews. Ethnographic methods can reveal systemic impacts of an AI system as it is adopted in institutions. For example, on AI systems other than generative AI, ethnographic methods revealed how embedding an AI system amplified unfair discrimination in a specific police force (Marda and Narayan, 2020). Ethnographic methods can also shed light on the functionality of an AI system at the point of use, i.e. an evaluation at the human interaction layer.

Pilot release studies may also constitute a safe environment for running experiments to isolate causal effects. For example, experimental roll-outs of AI systems have tested the impact on productivity of an AI system in the

workplace (Brynjolfsson et al., 2023). Human–AI interaction experiments as described above can also constitute a form of pilot study that targets systemic impact. This has been done to assess potential economic impacts, such as productivity impacts of co-writing with ChatGPT (Noy and Zhang, 2023) or developer productivity and happiness when co-programming with an AI (Peng et al., 2023). Other experimental approaches such as randomised controlled trials (RCTs) could also be applied to generative AI systems to identify impacts by comparing outcomes to organisations where AI systems are deployed differently or not at all.

Staged release and pilot studies have two main weaknesses. First, similarly to experimental methods, as described above, staged roll-outs or pilot projects may not allow for generalisations to novel contexts. This is particularly the case for highly specific studies on local contexts, such as ethnographic work. Small-scale experiments may also yield misleading results – for example, they may identify impacts that are negated by equilibrium effects that emerge through large-scale adoption (Lise et al., 2004). Further, pilot studies that have ecological validity regarding intended deployment contexts require some level of deployment of an AI system, which raises potential safety concerns. Ethnographic studies also require embedding researchers in potential deployment contexts, which can create disruption or discomfort for local institutions or systems.

Impact assessments

Impact assessments may be conducted before an AI system is deployed (e.g. algorithmic impact assessments (AIAs), Organisation for Economic Co-operation and Development (a); US Chief Information Officers Council), or retrospectively (also referred to as ex-post assessments). Impact assessments can track potential effects on broader economic, environmental, or social structures. Prospective impact assessments may be grounded in guiding questions to assess potential risks of harm. Retrospective impact assessments can be grounded in the monitoring of different system-level indicators and patterns. For example, observational data may monitor different economic indicators to quantify the impact on employment and inequality of AI-adjacent technologies such as robotics (Acemoglu and Restrepo, 2020; Acemoglu et al., 2020; Bonfiglioli et al., 2020). Retrospective impact assessments may also leverage expert views, for example, by surveying or interviewing groups that may witness systemic impacts, as has been done for other types of AI (Bell, 2023; Hunt et al., 2022). Interviewing different groups can further shed light on the distribution of systemic impacts from AI systems, such as one study finding increased negative emotions for low-skilled employees in contrast to increased creativity for high-skilled employees (e.g. Jia et al. (2023)). Instances of observed harm can be aggregated to identify patterns and systemic impacts (AI Incident Database). Impact assessments also comprise case studies, for example, evaluating the impact of generative AI systems on the creative commons, a public good (del Rio-Chanona et al., 2023; Huang and Siddarth, 2023), or impact on job automation (Milanez, 2023).

Impact assessments, differently from experiments and pilots, rely on qualitative and/or observational data about complex and prolonged processes, which makes it difficult to generate causal evidence of impact. This can be partially addressed by leveraging natural experiments such as policy shocks that introduce exogenous variation on technology adoption. However, this restricts the range of potential studies that can be undertaken and requires strong assumptions about the link between exogenous shock, technology adoption, and impact (Kiviet, 2020). Perhaps even more importantly, the evidence from ex-post assessment arrives too late to guide model development and deployment to prevent real-world harms or avoid potential lock-in to harmful technologies (Acemoglu and Lensman, 2023). Forecasts and experiments that can generate evidence of (potential) impact further upstream can help address this gap.

Forecasts and simulations

Forecasts and simulations of downstream impact can help evaluate risks of harm from AI systems on broader structures. For example, anticipated AI system capabilities have been mapped against tasks required to perform certain jobs, to forecast likely downstream impacts on labour markets (Autor et al., 2022; Frank et al., 2019; Webb, 2019). Comparative technologies can be evaluated as a proxy for potential impact from generative AI systems. Assessing their impact can help estimate impacts from a given AI system or serve as a starting point for determining what harms may arise and what mistakes should not be repeated (Autor et al., 2022; Webb, 2019). Where no highly similar technologies exist, loose analogies may be useful heuristics[15] but should be used with caution.

Forecasts rely on a range of ambiguous decisions, which lead to uncertain results. For example, in evaluations of economic exposure to automation based on analysis of the overlap between model capabilities and work tasks, much depends on how model capabilities are defined and quantified, making transparency in data sources and methods critical (Eloundou et al., 2023; Felten et al., 2021; Frank et al., 2019; Frey and Osborne, 2017; Tolan et al., 2021). By way of illustration, in the case of labour market forecasts, it is generally difficult to estimate if exposure will translate into a positive impact (e.g. a worker becomes more productive) or a negative one (a worker is displaced) without additional assumptions about the deployment context or ex-post analyses.

A.3. Case study: Misinformation

Misinformation is defined as the spread of false, inaccurate, or misleading information, often unintentionally (Wardle and Derakhshan, 2017).[16] Large multimodal AI systems can produce highly realistic but factually incorrect or misleading content, such as images of events that never took place or synthetic audiovisual representations of people saying things they never said ((Birnbaum and Davison, 2023; Weise and Metz, 2023). These are often described as "confabulations" or "hallucinations" when generated in response to an innocuous query (Ji et al., 2023; Li et al., 2023b; Xiao and Wang, 2021). False or misleading information can also be created and spread with the explicit intent to deceive – as part of large-scale disinformation campaigns, for example (Marwick and Lewis, 2017; Satariano and Mozur, 2023).

Factually inaccurate or fictitious outputs may be harmless, such as when used for creative or satirical purposes (Diaz et al., 2022; Kasirzadeh and Gabriel, 2023). However, in other contexts, misinformation generated from multimodal AI systems may pose a significant risk of harm, such as leading people to act on false beliefs, including in legal, medical, or other high-stakes contexts. People often struggle to reliably distinguish synthetic from human-generated content (Clark et al., 2021; Groh et al., 2022; Kreps et al., 2022; Nightingale and Farid, 2022; Spitale et al., 2023). Audiovisual misinformation spreads faster online and is perceived as more credible than text-based misinformation (Hameleers et al., 2020; Sundar et al., 2021), exacerbating potential downstream risks. In May 2023, for example, an AI-generated image that was widely circulated on Twitter led several news outlets to falsely report that an explosion had taken place at the US Pentagon, causing a brief drop in the US stock market (Alba, 2023).

AI-generated misinformation could also have broader societal repercussions, such as eroding public trust in evidence and information. As synthetic audiovisual content becomes more widespread, research suggests that people may become more uncertain about what to believe and, as a result, more distrustful of established sources of information (Lee et al., 2023d; Vaccari and Chadwick, 2020). A related concern is that inaccurate or misleading synthetic outputs created with generative AI systems could end up contaminating open "knowledge commons" (e.g. Wikipedia), jeopardising the quality and legitimacy of shared public knowledge (Huang

[15] For example, AI systems have been likened to other technological innovations such as steam engines and office automation (Acemoglu and Johnson, 2023).

[16] This is distinct from disinformation, which is false information created with the explicit intent to deceive or cause harm to others (Wardle and Derakhshan, 2017).